

Calculative Foundation

1. Dimensionality

Dimensionality refers to the number of features or variables in a dataset. Each feature represents one dimension (e.g., subjects like Math, Science). Higher dimensions mean more information but also more complexity.

2. Dimensionality Increase

Dimensionality increase means adding new features or transforming existing data into a higher-dimensional space. This is done by creating new variables such as totals, squares, or interaction terms. It helps capture more patterns in the data.

3. Feature Expansion

Feature expansion is the process of generating additional features from existing ones. For example, adding total marks or squared values of scores. This increases the dimensionality and improves data representation.

4. Polynomial Transformation

Polynomial transformation creates new features by raising variables to powers or combining them. For example, from x we create x^2 or xy . It helps model complex and non-linear relationships.

5. Line (1D Space)

A line represents one-dimensional space and is formed by scaling a single vector. It shows movement in only one direction. In data, it represents variation along one feature.

6. Plane (2D Space)

A plane is a two-dimensional space formed by two independent vectors. It allows representation of data using two features. Any point on the plane is a combination of these vectors.

7. Hyperplane (Higher Dimensions)

A hyperplane is a generalization of a plane in higher dimensions (3D and above). It separates data into different regions or classes. It is widely used in classification algorithms like SVM.

8. Curse of Dimensionality

As dimensions increase, data becomes sparse and harder to analyze. Distance calculations become less meaningful. This can reduce model performance if not handled properly.

9. Feature Interaction

Feature interaction occurs when two or more features combine to influence the output. For example, $\text{Math} \times \text{Science}$ marks. These interactions increase dimensionality and capture deeper relationships.

10. **Kernel Trick (Concept)**

The kernel trick is used to map data into higher dimensions without explicitly computing new features. It helps make data linearly separable. Commonly used in machine learning models like SVM.